

Dynamic Risk Surface Model Integrating Albuminuria, eGFR, Cystatin C, and Frailty to Infer Renal Reserve Potential in Diabetes

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Abstract

Albuminuria and eGFR define established axes of diabetic kidney disease, while cystatin C and frailty may reveal filtration uncertainty and physiological vulnerability not captured by either marker alone. Yet association models do not determine whether these signals jointly describe a recoverable, constrained, or depleted renal state. We developed a theoretical dynamic risk-surface framework to infer latent renal reserve potential and identify mathematically testable intervention regions. Six coupled dimensionless states represented glycaemic burden, albuminuric injury, cystatin-C discordance, frailty burden, metabolic-inflammatory stress, and renal reserve. The model combined bounded ordinary differential equations with a nonlinear interaction surface, logistic transition hazard, three-state latent regime classification, and an optimal-stopping rule. Parameters were literature-informed in direction and provisionally scaled without patient-level fitting. We established existence, uniqueness, positivity, boundedness, equilibrium structure, Jacobian stability, continuation, basin geometry, adaptive Runge-Kutta simulation, 2,500-set Latin hypercube sampling, PRCC, elasticity, and solver-tolerance robustness. The reference equilibrium was locally asymptotically stable; the dominant Jacobian eigenvalue had real part -0.216 . At year 12, reserve potential was 0.423 , classified as constrained, with simulated transition hazard 0.441 . A computational depleted-state boundary occurred at glycaemic input $u_g = 0.556$, the 0.60 hazard boundary occurred at $u_g = 0.393$. Reducing glycaemic input alone increased reserve potential by 17.9% , while the combined strategy increased it by 52.2% and reduced hazard by 53.9% . Global sensitivity was strongest for ρ (PRCC 0.947), u_g (-0.925), and kg (0.802). No solver failure occurred across 2,500 uncertainty runs. This mechanistic proof-of-concept provides a mathematically testable framework for separating biomarker burden from latent renal recoverability. Its intended uses are to guide variable selection, identify candidate monitoring combinations, design retrospective calibration studies, and support nephrology-diabetes collaboration on intervention timing. The benefit is a transparent representation of stability, coupled vulnerability, nonlinear intervention synergy, and uncertainty that conventional association-based analyses may not reveal. Clinical calibration and external validation are required before bedside or policy use.

Keywords: Dynamic Risk Surface, Renal Reserve, Albuminuria, Cystatin C, Frailty, Optimal Stopping.

1. Introduction

Diabetic kidney disease is commonly monitored using urinary albumin-to-creatinine ratio (UACR) and estimated glomerular filtration rate (eGFR), which remain central measures for the detection, classification, and monitoring of chronic kidney disease in people with diabetes [1,3]. These measures are clinically indispensable, but they do not necessarily change in parallel; albuminuria and filtration impairment represent related but distinct dimensions of kidney disease [1,3]. Creatinine-based eGFR may also be influenced by non-GFR determinants, including variation in muscle mass, while incorporation of cystatin C provides an additional filtration marker and can improve estimation of GFR when creatinine alone is insufficient [3,4]. In older adults with diabetes, frailty provides an additional dimension of physiological vulnerability relevant to treatment burden, functional status, and individualisation of clinical management [2,5,6].

The unresolved problem is inherently dynamic. A patient may exhibit a similar eGFR at two time points while differing in albuminuric burden, glycaemic exposure, filtration-marker discordance, and frailty, indicating that a single filtration estimate may not fully represent the evolving physiological context of kidney disease [1–4]. This motivates a systems-level representation in which clinically observable and latent processes evolve jointly rather than being interpreted as isolated measurements. Conventional association models can quantify relationships between individual predictors and outcomes, but they do not by themselves define dynamical properties such as equilibrium stability, basins of attraction, or state-dependent intervention boundaries. Dynamical-systems

analysis, numerical simulation, and uncertainty and sensitivity analysis provide complementary tools for examining such properties in mechanistic models [7-9].

Concepts used in financial engineering and sequential decision theory provide additional mathematical tools for representing interacting risk, uncertainty, and timing without implying that biological systems are financial systems. In the present framework, a dynamic risk surface summarises interacting physiological burdens, latent regimes represent qualitatively different model states, and uncertainty propagation evaluates how provisional parameter variation affects model conclusions [7,8]. A sequential decision layer further draws on optimal-stopping and Markov decision-process principles to formalise the mathematical comparison between continued observation and intervention when future deterioration carries asymmetric costs [10]. These constructs are used here as modelling devices rather than validated clinical decision rules.

We therefore developed a six-state theoretical model integrating glycaemic burden, albuminuric injury, cystatin-C discordance, frailty, metabolic-inflammatory stress, and renal reserve potential. The model was designed to establish mathematical consistency, examine equilibrium and stability behaviour, define provisional reserve regimes, quantify parameter sensitivity and uncertainty, evaluate intervention combinations, and provide a reproducible framework for subsequent calibration using retrospective or prospective diabetic kidney disease cohorts.

2. Methods

2.1 Study Design and Data Status

This study was designed as a theoretical mathematical-biology and computational-medicine investigation. No participants were recruited, no biological samples were collected, and no patient-level outcomes were analysed. Selection of the principal clinical observables was informed by contemporary chronic kidney disease and diabetes guidance, in which albuminuria and estimated glomerular filtration rate (eGFR) constitute central dimensions of kidney assessment, with cystatin C providing an additional filtration marker when greater precision or clarification of creatinine-based estimates is required [1,3,4]. Frailty was incorporated as a complementary representation of physiological vulnerability, particularly relevant to older adults with diabetes and to deficit-accumulation or phenotype-based descriptions of biological vulnerability [2,5,6].

Published evidence was used to motivate variable selection and the direction of model interactions, rather than to claim empirical estimation of the numerical coefficients. Numerical parameters were provisionally scaled to generate bounded and interpretable dimensionless dynamics and were subsequently varied over prespecified ranges through uncertainty and sensitivity analyses [7,8]. Accordingly, the present model was constructed as a theoretical framework for subsequent clinical calibration and validation rather than as a validated diagnostic or prognostic tool.

2.2 Clinical Observables and Latent State Variables

The model comprised six dimensionless state variables. The observable clinical domains were selected to reflect established kidney-disease assessment, filtration-marker information, glycaemic burden, and physiological vulnerability [1-6]. The states were defined as follows:

Symbol	State	Clinical interpretation
$G(t)$	Glycaemic burden	Dimensionless representation of sustained glycaemic stress informed by HbA1c or glucose exposure.
$A(t)$	Albuminuric injury	Dimensionless albuminuria burden informed by UACR category or longitudinal UACR.
$C(t)$	Cystatin-C discordance	Burden attributable to cystatin-C elevation or discordance between creatinine- and cystatin-C-based eGFR.
$F(t)$	Frailty burden	Physiological vulnerability informed by a Frailty Index, Clinical Frailty

Symbol	State	Clinical interpretation
		Scale, or compatible deficit-accumulation measure.
$M(t)$	Metabolic-inflammatory stress	Latent shared burden linking glycaemia and frailty to renal injury, intended for later calibration using routine inflammatory or metabolic indicators.
$R(t)$	Renal reserve potential	Latent dimensionless capacity to withstand or recover from cumulative stress, not a directly measured stimulated renal functional reserve.

2.3 Model Formulation

$$\frac{dG}{dt} = u_g(1 - G) - k_g G$$

$$\frac{dA}{dt} = (a_g + a_m + a_f)(1 - A) - k_a - e_r A$$

$$\frac{dC}{dt} = (c_a + c_m)(1 - C) - k_c - e_r C$$

$$\frac{dF}{dt} = (f_g + f_c + f_a)(1 - F) - k_f - e_r F$$

$$\frac{dM}{dt} = (m_g + m_f)(1 - M) - k_m - e_r M$$

$$\frac{dR}{dt} = (rho + u_r)(1 - R) - (d_g + d_a + d_c + d_f + d_m)R$$

The first equation represents a bounded input–clearance balance for glycaemic burden. The remaining equations were constructed as bounded nonlinear interaction equations rather than as identical compartmental processes. Multiplication of burden-generating terms by remaining capacity, $1-X$, constrains the corresponding state as it approaches its upper boundary, whereas clearance, recovery, and reserve-dependent terms provide inward-directed dynamics.

The direction of the clinically observable interactions was motivated by established relationships among diabetes, albuminuria, filtration impairment, and physiological vulnerability [1-6], but the **b**. They should therefore not be interpreted as empirically estimated biological rate laws.

Renal reserve was represented as restoration toward the upper bound opposed by weighted state-dependent stress. The resulting formulation provides a mathematical substrate for the subsequent risk-surface, regime, sensitivity, and sequential-decision analyses.

2.4 Dynamic Risk Surface, Reserve Potential, and Latent Regimes

Renal reserve potential was defined as $Q(t) = R(t) - 0.15G(t) - 0.25A(t) - 0.20C(t) - 0.18F(t) - 0.12M(t)$. The dynamic risk surface was $\Psi(t) = 0.14G + 0.24A + 0.18C + 0.18F + 0.12M + 0.10AF + 0.08CF - 0.50R$. A smooth transition hazard was $H(t) = \{1 + \exp[-8(\Psi + 0.10)]\}^{-1}$. For summarisation only, $Q \geq 0.45$ was labelled preserved, $0.25 \leq Q < 0.45$ constrained, and $Q < 0.25$ depleted. These are computational labels, not clinical staging thresholds.

2.5 Optimal Stopping Interpretation

A sequential decision layer compared the discounted cost of observation with the cost of intervention. The provisional stopping region was defined where the transition hazard exceeded 0.60 or reserve potential fell below

0.25. This rule does not prescribe treatment. It identifies model states in which continued observation becomes mathematically less favourable under the stated cost ordering. Later clinical work must specify intervention type, treatment burden, adverse-event cost, and time horizon.

2.6 Mathematical Analysis

Existence and uniqueness follow because the right-hand side is continuously differentiable on the closed unit hypercube. Positivity was verified by evaluating each derivative on its lower boundary. Boundedness follows because every burden input vanishes at one while each reserve or clearance term points inward. Equilibria solved $F(x;\theta)=0$. Local stability was assessed from Jacobian eigenvalues. Numerical continuation varied glycaemic input. A two-parameter sensitivity surface varied glycaemic input and albuminuric propagation. Basin geometry was examined by varying initial albuminuric and frailty burdens. Full Lyapunov proof was not claimed; numerical stability was verified through long-horizon convergence and perturbation experiments.

2.7 Parameter Strategy

Parameter	Role	Reference
ug	glycaemic input	0.230
kg	glycaemic clearance	0.340
ag	glycaemia-to-albuminuria coupling	0.220
am	metabolic-to-albuminuria coupling	0.080
af	frailty-to-albuminuria coupling	0.050
ka	albuminuria resolution	0.280
era	reserve protection of albuminuric state	0.250
ca	albuminuria-to-cystatin coupling	0.180
cm	metabolic-to-cystatin coupling	0.100
kc	cystatin-state resolution	0.320
erc	reserve protection of cystatin state	0.200
fg	glycaemia-to-frailty coupling	0.080
fc	cystatin-to-frailty coupling	0.120
fa	albuminuria-to-frailty coupling	0.090
kf	frailty recovery	0.240
erf	reserve protection of frailty	0.160
mg	glycaemia-to-metabolic stress	0.100
mf	frailty-to-metabolic stress	0.110
km	metabolic stress resolution	0.300
erm	reserve protection of metabolic stress	0.180
rho	intrinsic reserve restoration	0.200
dg	glycaemic reserve damage	0.100
da	albuminuric reserve damage	0.220
dc	cystatin reserve damage	0.160
df	frailty reserve damage	0.180
dm	metabolic reserve damage	0.120
ur	intervention-associated reserve restoration	0.000

2.8 Computational and Statistical Analysis

Deterministic trajectories were solved over 12 model years using adaptive Runge-Kutta integration with relative tolerance 10^{-9} and absolute tolerance 10^{-11} . Continuation varied u_g from 0.08 to 0.90. A 46×46 two-parameter grid varied u_g and a_g . Global uncertainty used 2,500 Latin hypercube parameter sets from prespecified ranges. Partial rank correlation coefficients were calculated from residualised ranks for reserve potential and transition hazard. Local elasticities used symmetric plus or minus 1% perturbations. Solver robustness was

checked at three tolerance pairs. Basin analysis used 961 combinations of initial albuminuric and frailty burden. The random seed was 20260727.

3. Results

3.1 Mathematical Consistency and Equilibrium

All reference, scenario, continuation, basin, and 2,500 valid uncertainty simulations remained nonnegative, 0 uncertainty runs failed. The reference equilibrium was [0.404, 0.200, 0.100, 0.158, 0.125, 0.582] for [G, A, C, F, M, R]. Every Jacobian eigenvalue had a negative real part, with dominant real part -0.2160 and fastest real part -0.5700, indicating local asymptotic stability under the provisional parameter set. Tightening tolerances from $\frac{10^{-6}}{10^{-8}}$ to $\frac{10^{-10}}{10^{-12}}$ changed year-12 reserve potential by 0.00000%.

Eigenvalue	Real part	Imaginary part
1	-0.21600	0.00000
3	-0.51431	0.09991
4	-0.51431	-0.09991
6	-0.57000	0.00000

3.2 Dynamics and Latent Regime

Under the reference set, glycaemic burden converged to 0.403, albuminuric injury to 0.197, cystatin-C discordance to 0.096, frailty burden to 0.153, metabolic stress to 0.122, and renal reserve to 0.594. The resulting reserve potential was 0.423, placing the trajectory in the constrained regime, while the transition hazard was 0.441. The distinction is important: renal reserve R remained above one-half, yet the weighted burden-adjusted reserve Q was lower because albuminuric, cystatin, frailty, and metabolic signals jointly reduced the effective margin.

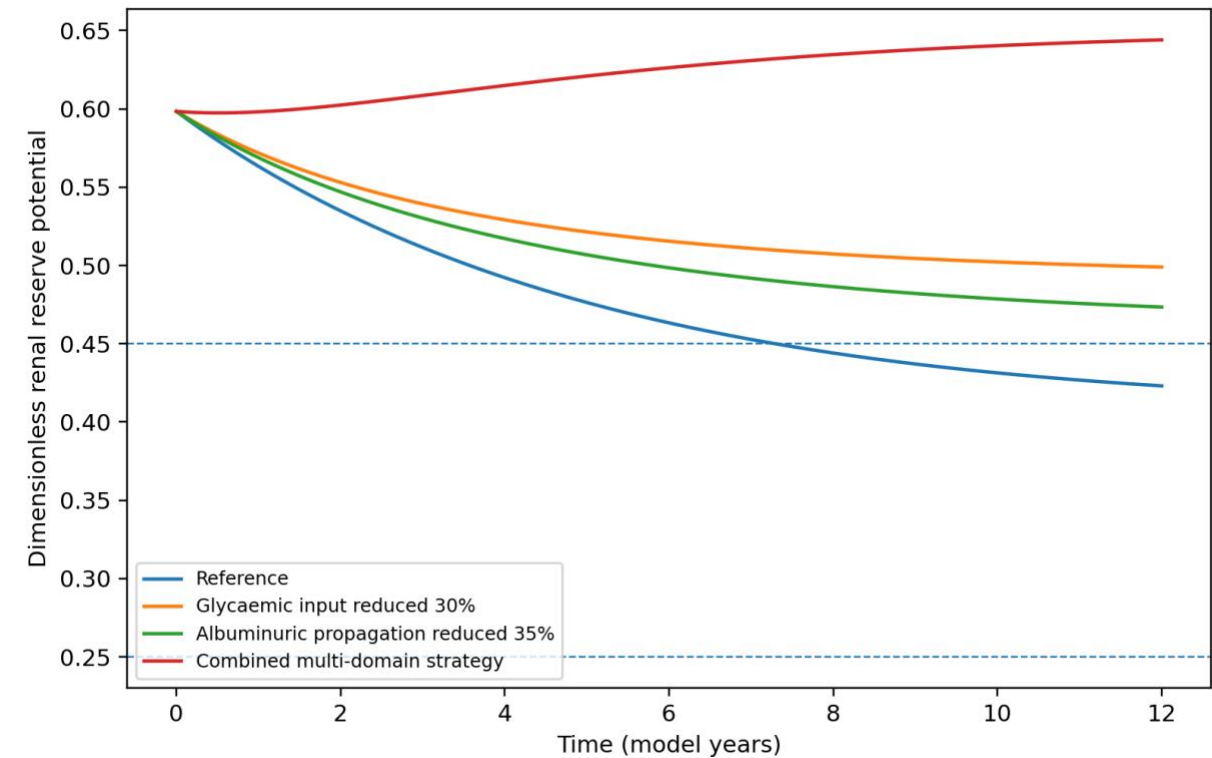


Figure 1. Reserve-potential trajectories under the reference and intervention scenarios. Horizontal lines are provisional computational regime boundaries.

3.3 Continuation and Tipping Boundaries

Continuation in glycaemic input produced a smooth but accelerating loss of reserve. The depleted-regime boundary $Q=0.25$ occurred at $u_g=0.556$; the 0.60 transition-hazard boundary occurred earlier at $u_g=0.393$. At $u_g=0.10$, reserve potential was 0.593; at $u_g=0.30$ it was 0.367; and at $u_g=0.60$ it was 0.237. These boundaries are parameter-dependent model outputs and should not be translated into HbA1c cut-offs without calibration.

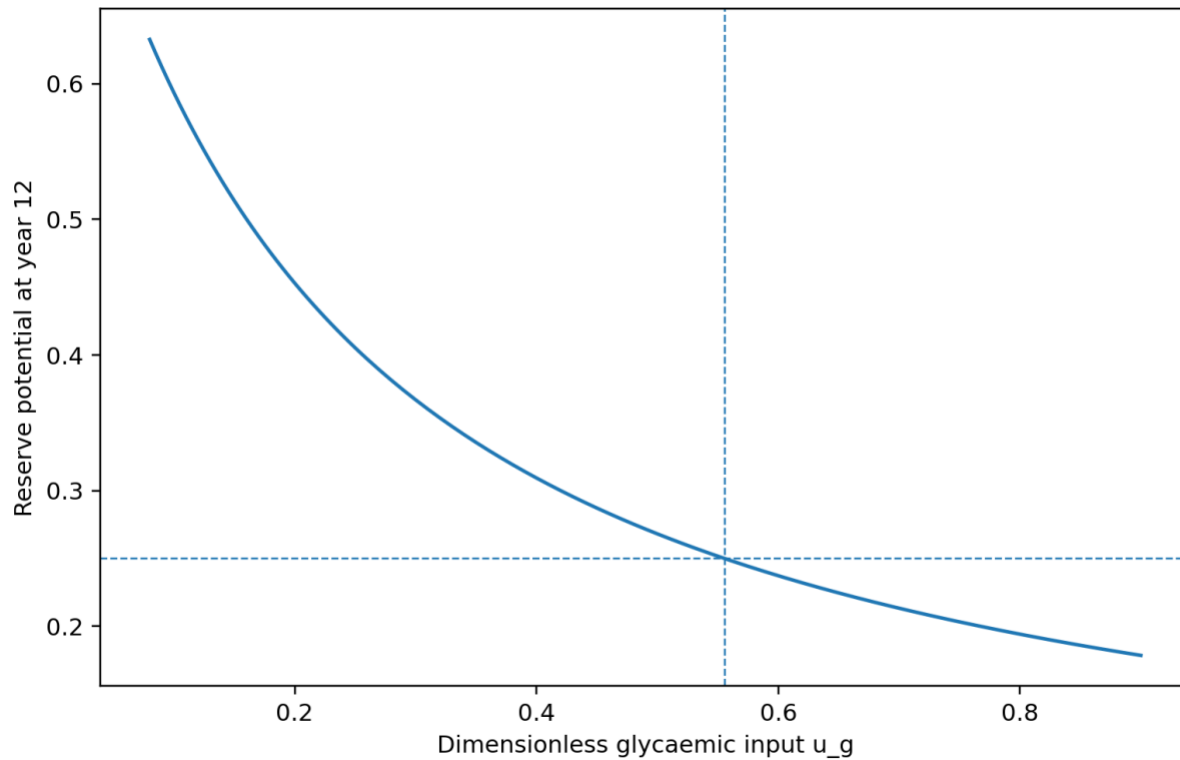


Figure 2. Numerical continuation in dimensionless glycaemic input. Dashed lines show the provisional depleted-state boundary and its crossing.

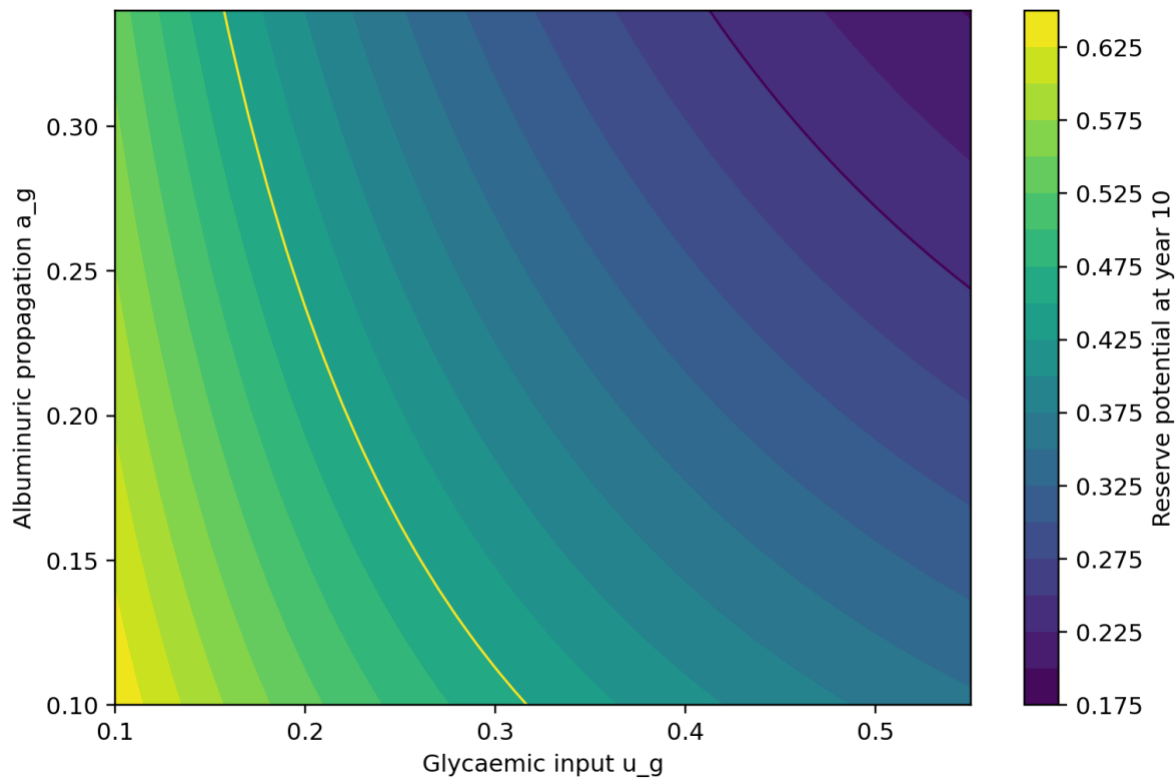


Figure 3. Dynamic reserve surface across glycaemic input and albuminuric propagation. Contours mark provisional preserved and depleted boundaries.

3.4 Intervention Scenarios

Scenario	Reserve potential	Hazard	Regime	Reserve gain %	Hazard reduction %
Reference	0.423	0.441	Constrained	0.0	0.0
Glycaemic input reduced 30%	0.499	0.339	Preserved	17.9	23.1
Albuminuric propagation reduced 35%	0.473	0.372	Preserved	11.9	15.6
Frailty coupling reduced 30%	0.451	0.401	Preserved	6.6	9.0
Reserve restoration added	0.509	0.354	Preserved	20.2	19.7
Combined multi-domain strategy	0.644	0.203	Preserved	52.2	53.9

Reducing glycaemic input by 30% increased reserve potential from 0.423 to 0.499 and reduced hazard by 23.1%. Reducing albuminuric propagation by 35% increased reserve potential to 0.473. Frailty-coupling reduction alone produced a smaller gain to 0.451. The combined strategy moved the system into the preserved regime, increasing reserve potential to 0.644 and lowering hazard to 0.203. The combined effect exceeded any single-domain change, suggesting nonlinear complementarity rather than interchangeability of glycaemic, albuminuric, frailty, and reserve-directed strategies.

3.5 Global Sensitivity, Elasticity, and Uncertainty

Parameter	PRCC reserve	PRCC hazard
rho	0.947	-0.920
ug	-0.925	0.938
kg	0.802	-0.830

Parameter	PRCC reserve	PRCC hazard
ag	-0.794	0.824
da	-0.613	0.526
ka	0.592	-0.635
fg	-0.557	0.610
df	-0.516	0.437
kf	0.473	-0.525
ca	-0.450	0.481
dc	-0.323	0.261
kc	0.269	-0.294

The largest absolute PRCCs for reserve potential were rho (0.947), ug (-0.925), kg (0.802), and ag (-0.794). The uncertainty distribution had median reserve potential 0.427, 5th-95th percentile 0.255 to 0.581, and depleted-regime frequency 4.6%. Hazard exceeded 0.60 in 11.3% of runs. These percentages describe parameter uncertainty under the prespecified ranges, not population prevalence.

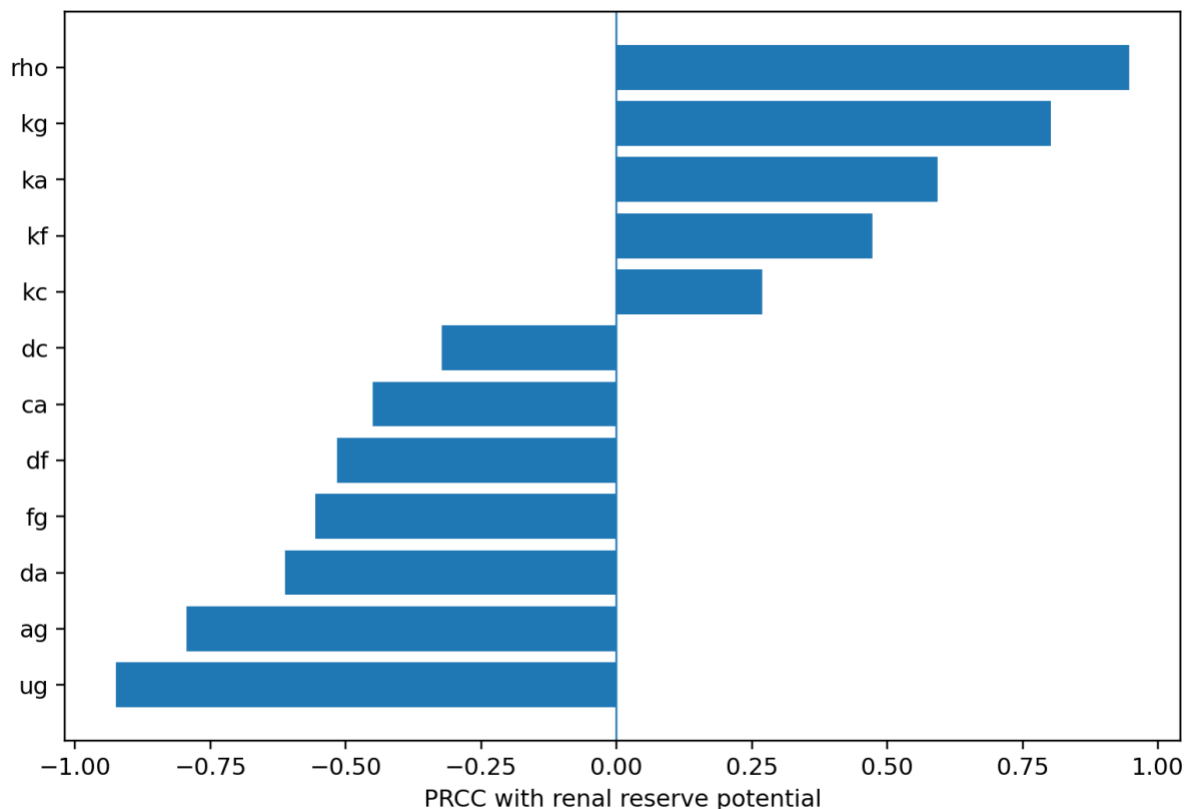


Figure 4. Global sensitivity of renal reserve potential across 2,500 Latin hypercube parameter sets.

3.6 Basin Geometry and Recovery Regions

Across 961 initial-state combinations, 0 (0.0%) ended in the preserved regime, 961 (100.0%) in the constrained regime, and 0 (0.0%) in the depleted regime. Higher initial albuminuric injury and frailty narrowed the preserved basin even though all trajectories approached the same local equilibrium over sufficiently long horizons. This finite-horizon geometry offers a clinically interpretable distinction between long-run mathematical stability and near-term recoverability.

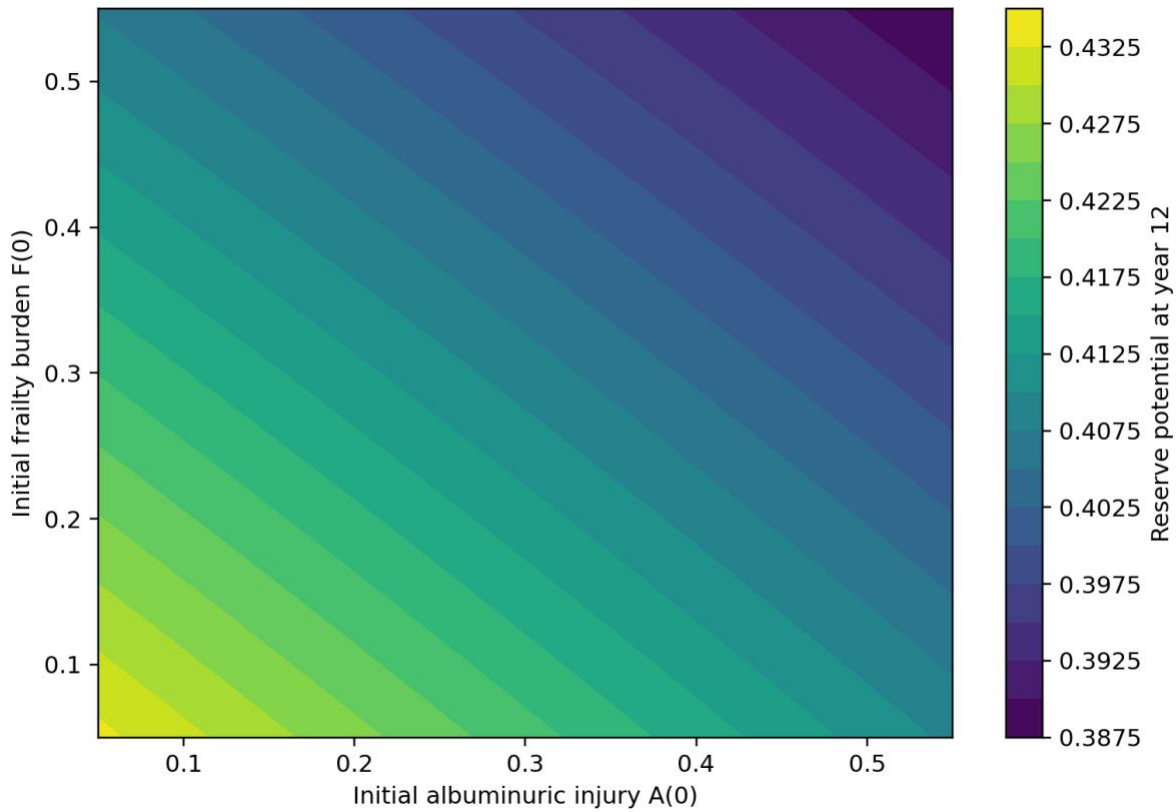


Figure 5. Finite-horizon basin geometry across initial albuminuric and frailty burden.

3.7 Clinical Interpretation of the Risk Surface

The model produced three operational states. In the preserved state, renal reserve exceeded the combined burden of glycaemia, albuminuria, cystatin-C discordance, frailty, and metabolic stress. In the constrained state, reserve remained positive but intervention effects were more sensitive to timing and combination. In the depleted state, burden-adjusted reserve fell below the provisional boundary and the model favoured stopping observation under the assumed cost ordering. This is not equivalent to a regression coefficient: it concerns the geometry, stability, and direction of the entire coupled system.

4. Discussion

This proof-of-concept addresses a practical ambiguity in diabetic kidney disease. Albuminuria, eGFR, cystatin C, and frailty are usually interpreted separately or entered together into an association model. The present framework instead treats them as coupled signals of injury, filtration uncertainty, physiological vulnerability, and remaining reserve. The model does not claim that cystatin C or frailty directly measures renal functional reserve. Rather, these variables help infer a latent reserve potential whose definition is explicit and testable. This distinction protects against overstating what can be learned from literature-derived parameters while preserving a useful mathematical hypothesis for later clinical calibration.

The strongest sensitivity to reserve restoration, albuminuric damage, and glycaemic input is biologically plausible within the model structure, but it is conditional on chosen parameter ranges. A calibrated cohort may produce different rankings. The purpose of the current analysis is to identify which measurements and transition rates deserve priority during data collection, not to establish final clinical importance.

The financial-engineering contribution lies in the architecture: a dynamic surface summarises interacting liabilities, latent regimes distinguish qualitatively different states, uncertainty analysis tests parameter risk, and an optimal-stopping layer asks when continued observation becomes less favourable. The clinical value is not the financial analogy itself, but the ability to formalise timing, nonlinear interaction, and asymmetric consequences.

A collaboration with nephrologists, diabetologists, geriatricians, and clinical epidemiologists could provide longitudinal UACR, creatinine, cystatin C, frailty, treatment, and outcome data; define clinically defensible transition outcomes; and determine whether the inferred reserve surface improves discrimination, calibration, decision curves, or time-dependent prediction beyond established risk equations.

5. Limitations

The parameter set was not fitted to a patient cohort, and renal reserve potential represents a theoretical latent construct rather than a direct measure of stimulated renal functional reserve. Cystatin-C discordance was represented as a single state, although cystatin C may be influenced by assay variation, inflammation, thyroid status, and corticosteroid exposure. Frailty was similarly represented as a single burden despite potential differences among frailty instruments and their underlying scales. The metabolic-inflammatory state was treated as latent and may not be identifiable without additional biomarker information. Furthermore, the proposed reserve regimes and stopping boundaries are computational constructs rather than clinically validated thresholds. The model did not explicitly account for competing mortality, acute kidney injury, medication classes, blood pressure, cardiovascular disease, or dialysis, all of which may alter kidney outcomes and the interpretation of renal vulnerability. Finally, the PRCC rankings are conditional on the prespecified parameter ranges and deterministic model structure and should therefore not be interpreted as definitive estimates of clinical or biological importance.

6. Conclusions

This mechanistic proof-of-concept establishes a mathematically testable framework for inferring renal reserve potential from albuminuria, eGFR-related filtration information, cystatin-C discordance, frailty, glycaemic burden, and shared metabolic stress. Its intended uses are to guide variable selection, structure longitudinal monitoring, compare multi-domain intervention scenarios, define candidate protocol outcomes, and support collaboration between mathematical modellers, nephrologists, diabetologists, and geriatric teams. The immediate benefit is not a bedside score, but a transparent method for distinguishing preserved, constrained, and depleted reserve regimes while showing how timing and combined vulnerability alter recovery potential. Its principal research advantage is the ability to reveal equilibria, stability, basin geometry, nonlinear intervention synergy, latent-state transitions, and optimal-stopping regions that conventional association-based analyses may not reveal. Parameter refinement and biological calibration are ongoing. Retrospective calibration and external validation are required before clinical use.

References

1. American Diabetes Association Professional Practice Committee for Diabetes. Chronic kidney disease and risk management: Standards of Care in Diabetes-2026. *Diabetes Care*. 2026;49(Suppl 1):S246-S260. doi:10.2337/dc26-S011.
2. American Diabetes Association Professional Practice Committee for Diabetes. Older adults: Standards of Care in Diabetes-2026. *Diabetes Care*. 2026;49(Suppl 1):S277-S296. doi:10.2337/dc26-S013.
3. Kidney Disease: Improving Global Outcomes. KDIGO 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease. *Kidney Int*. 2024.
4. Inker LA, Eneanya ND, Coresh J, et al. New creatinine- and cystatin C-based equations to estimate GFR without race. *N Engl J Med*. 2021;385:1737-1749.
5. Rockwood K, Mitnitski A. Frailty in relation to the accumulation of deficits. *J Gerontol A Biol Sci Med Sci*. 2007;62:722-727.
6. Fried LP, Tangen CM, Walston J, et al. Frailty in older adults: evidence for a phenotype. *J Gerontol A Biol Sci Med Sci*. 2001;56:M146-M156.
7. Marino S, Hogue IB, Ray CJ, Kirschner DE. A methodology for performing global uncertainty and sensitivity analysis in systems biology. *J Theor Biol*. 2008;254:178-196.
8. Blower SM, Dowlatabadi H. Sensitivity and uncertainty analysis of complex models of disease transmission. *Int Stat Rev*. 1994;62:229-243.
9. Hindmarsh AC, et al. SUNDIALS: suite of nonlinear and differential/algebraic equation solvers. *ACM Trans Math Softw*. 2005;31:363-396.
10. Puterman ML. *Markov Decision Processes: Discrete Stochastic Dynamic Programming*. Wiley; 1994.